LAB: System & Network and Administration

Course code: ITC-304L

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Semester: 5th

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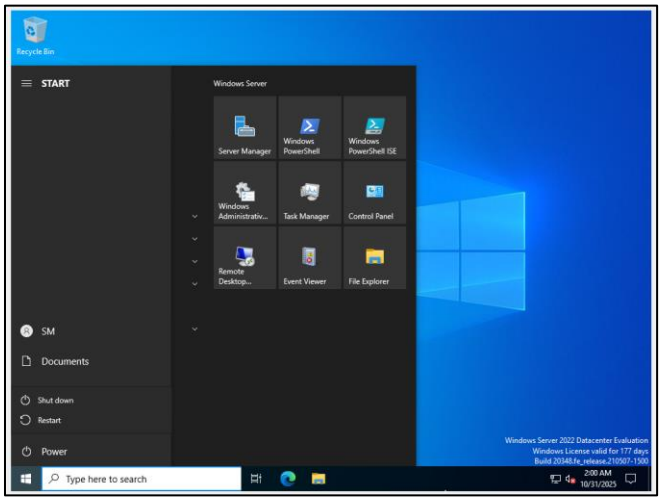
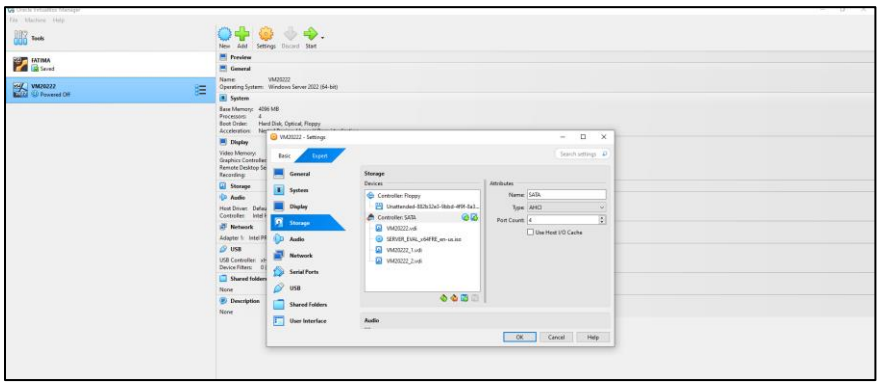
RAID SYSTEM

♦ Applying RAID System :

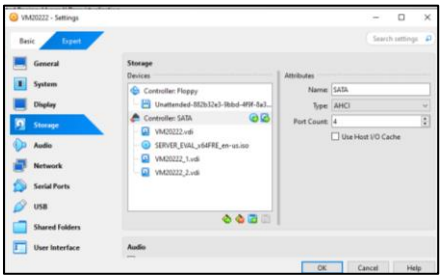
- To apply **RAID System** for Creating disk, at first ***shut down the VM machine*** as shown below.



- Then open the ***VM machine settings***.

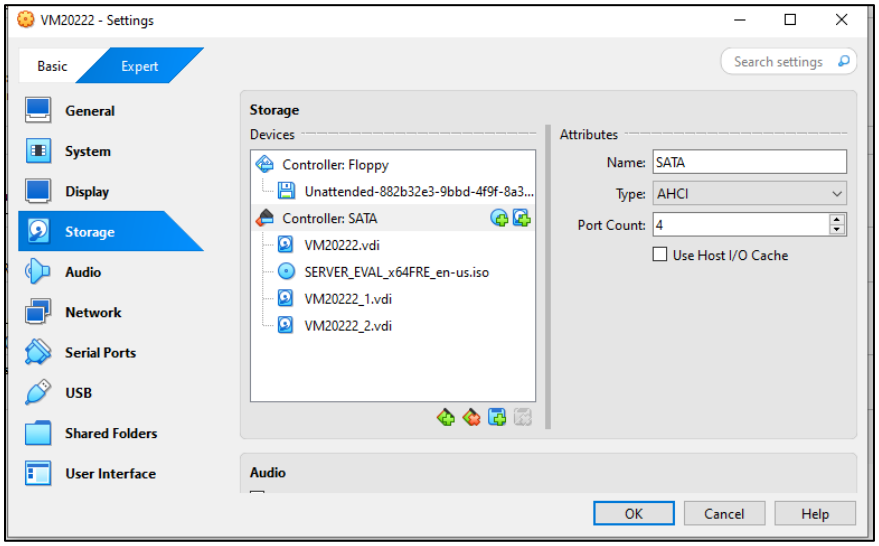


- In ***settings find the storage bar at left side option***.

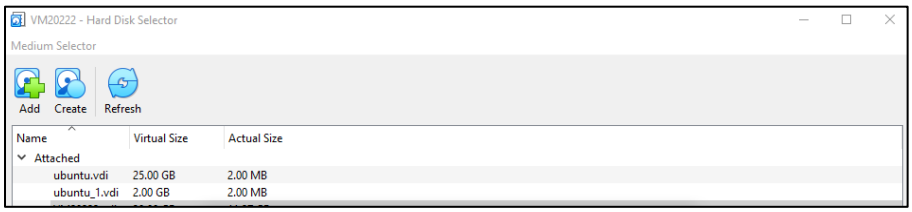


- In the storage bar there are two icons on ***right side of the Controller: SATA.***

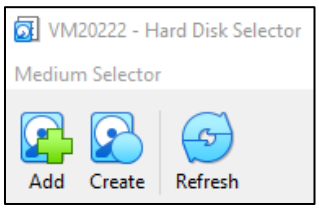
- Click on ***second icon.***



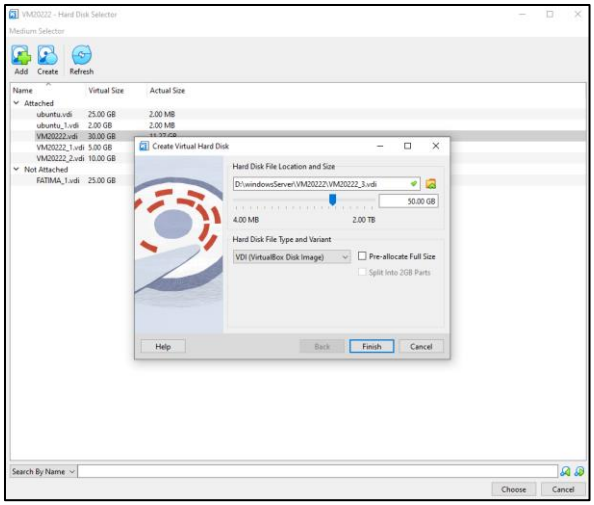
- After press the Add hard disks icon this screen shown below will appear.



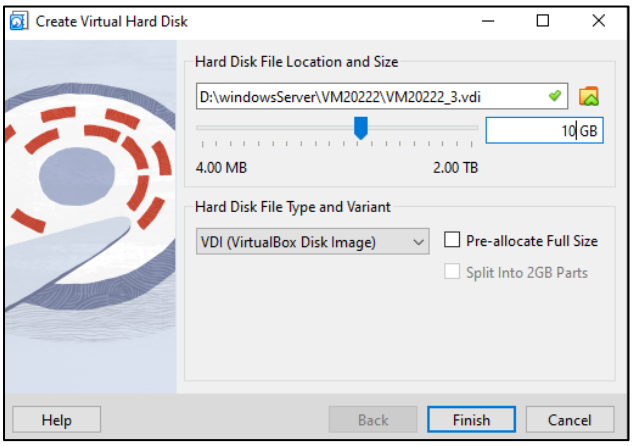
- In this screen click on ***Create button*** will be show at top.



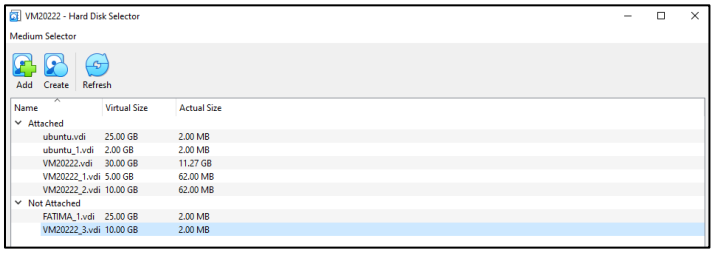
- After press create button than Create Virtual Hard Disk screen will appear.



- In this screen ***Change size of Hard Disk*** like 10GB, then press ***Finish***.

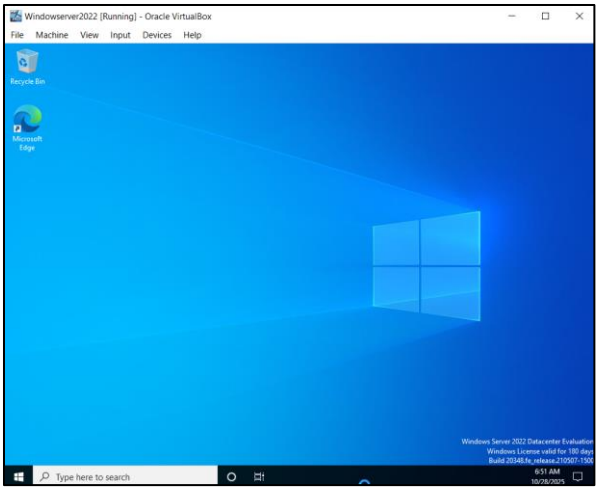


- After press Finish, In this screen new created disk 3 is in ***Not attached portion***.

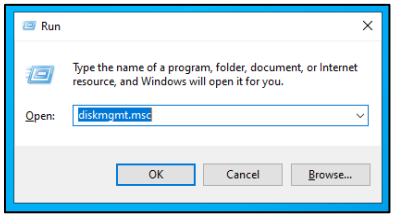


- To move it in attached click on it & press ***choose*** option as shown in below image.

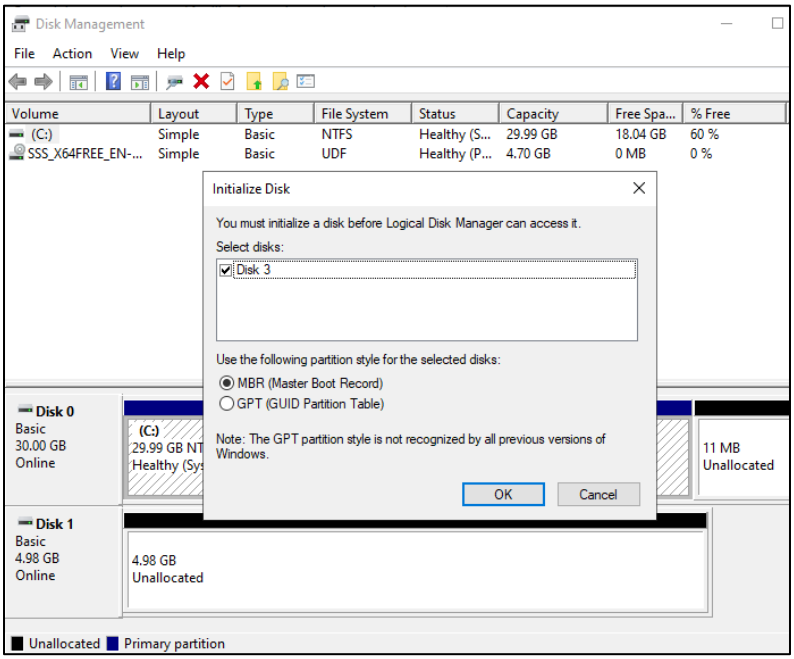
- Now ***Start*** the VM machine.



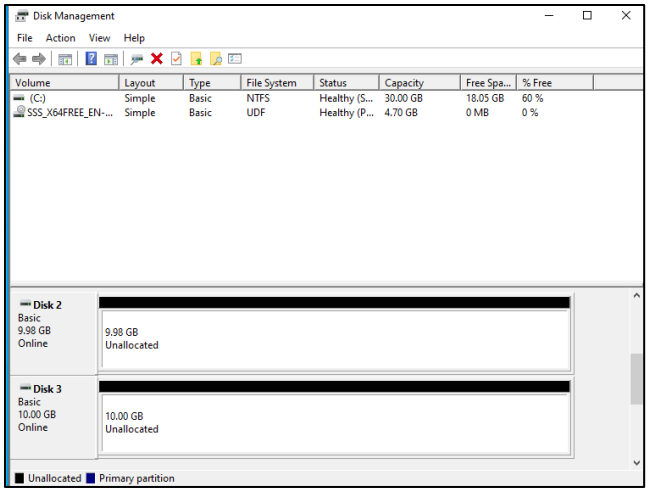
- When the desktop of VM machine appear press ***Win+R*** from keyboard & write ***diskmgmt.msc*** then press **OK.**



- After pressing OK the screen appears & now the new screen is ***initialize Disk***. Select ***MBR*** & press ***Ok***.



- ***Disk 3*** is created successfully.

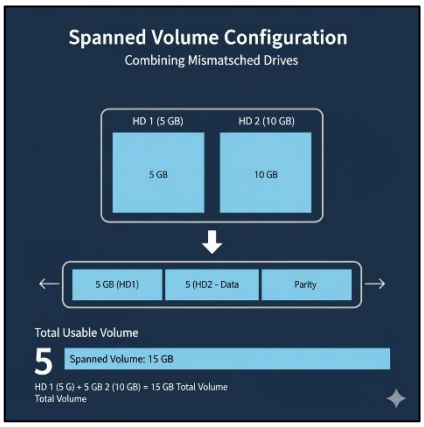


Remember: RAID system needs multiple disks for working we can create other disks by same method explained above for Disk 3.

♦ Applying RAID 0:

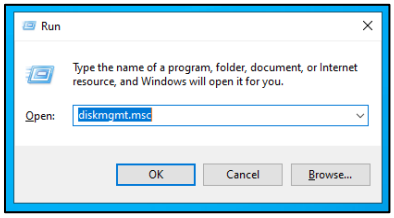
i) Spanned Volume (Working)

Feature	Details
Working	A Spanned Volume chains disks together to create one big storage space. Data is written sequentially: it fills the first disk completely before starting to use the second disk.
Goal	Maximum total capacity.
Result	Slow performance (like one disk) and zero protection (if one disk fails, all data is lost).
Example	HD 1 (5 GB) + HD 2 (10 GB) = 15 GB Spanned Volume.

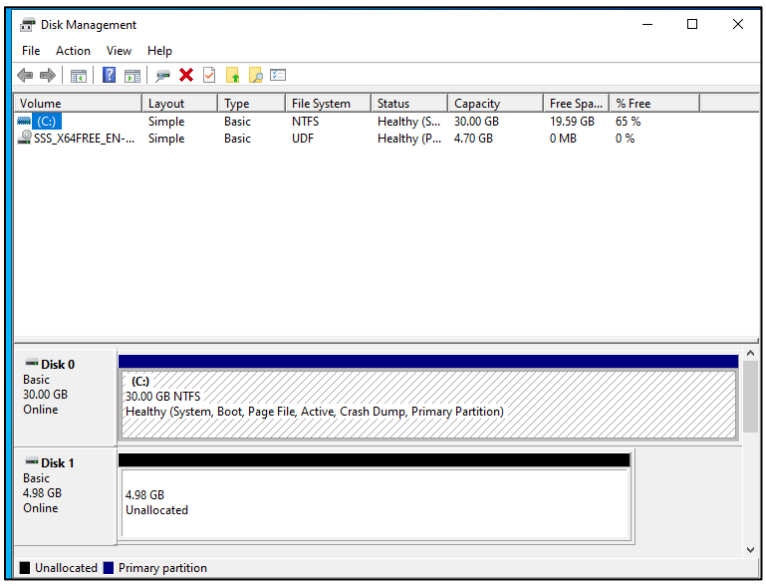


Apply Spanned method: For applying ***RAID 0 spanned method*** there must be two disks. Now we will apply spanned method on ***Disk 1 & Disk 2***. Disk 1 contains 5GB & Disk 2 contains 10GB.

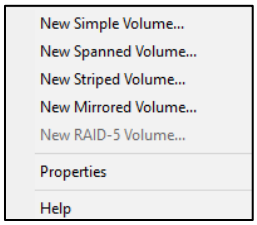
- Then start VM machine press ***Win+R*** & type ***diskmgmt.msc*** then press **OK.**



- Press **OK** then this screen will be appears in which ***disk 1*** is showing**.**

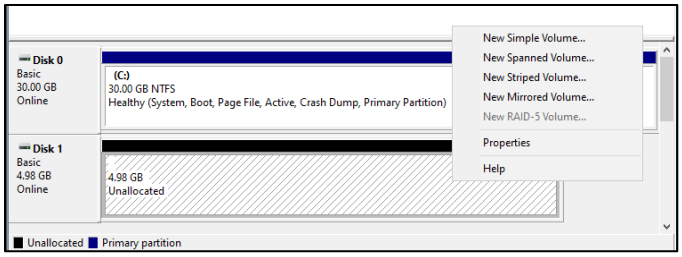


- ***Right Click on Disk 1***, then different options will be appears.

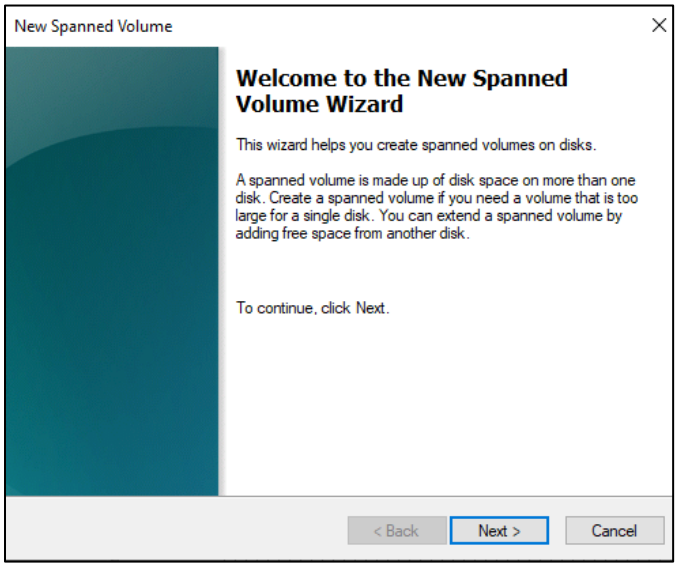


In screen ***New RAID-5 Volume is not available*** because we have only ***two disks Disk 1 & Disk 2*** & ***RAID 5 requires three disks***.

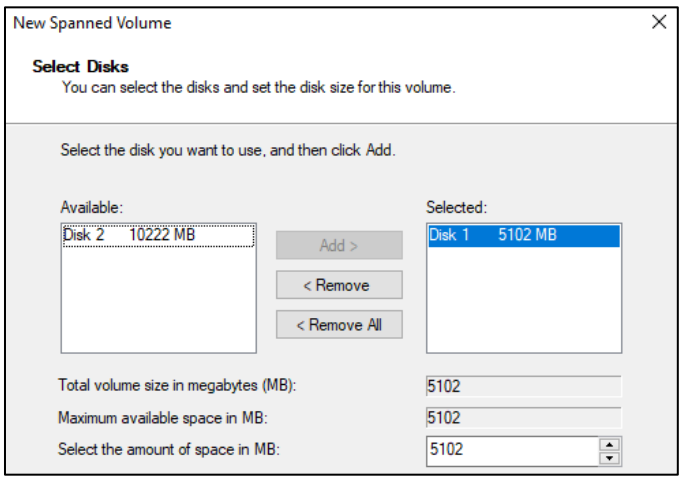
- Choose ***New Spanned volume*** *by click on it.*



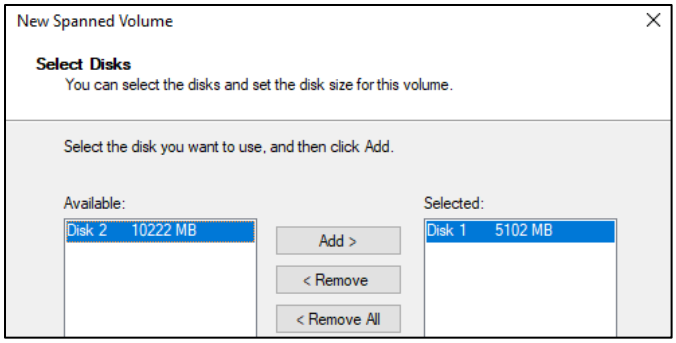
- This screen will be appear press ***Next***.



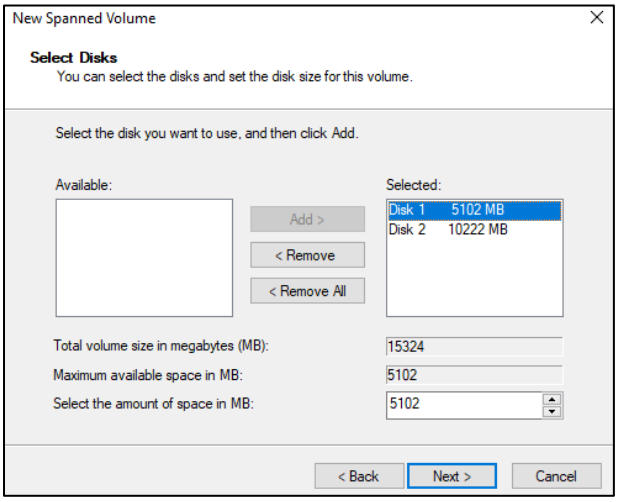
- In this screen we have two disks, ***Disk 2*** on left side is not highlight. ***Disk 1*** on Right side highlighted.



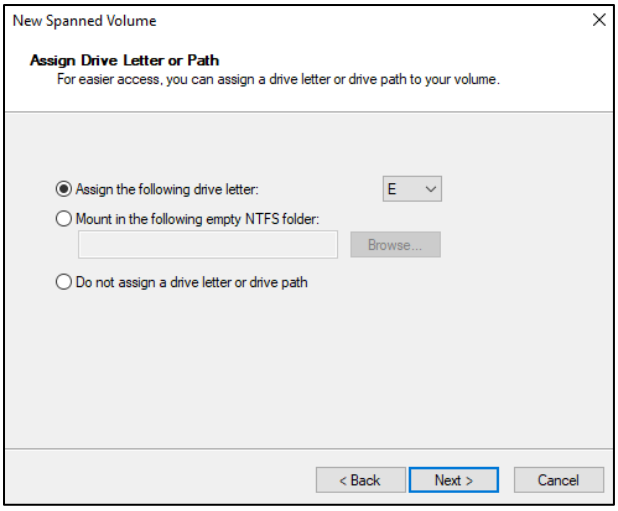
- Now ***select Disk 2*** & Press ***ADD.***



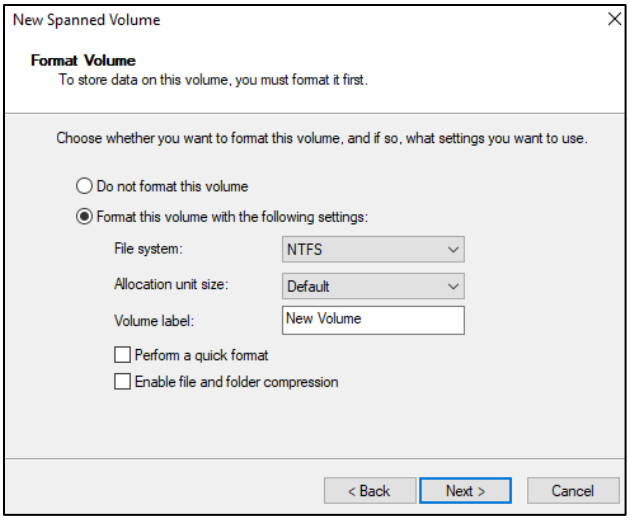
- Then ***Disk 2 is showing in the selected area below Disk 1***. Press ***Next***.



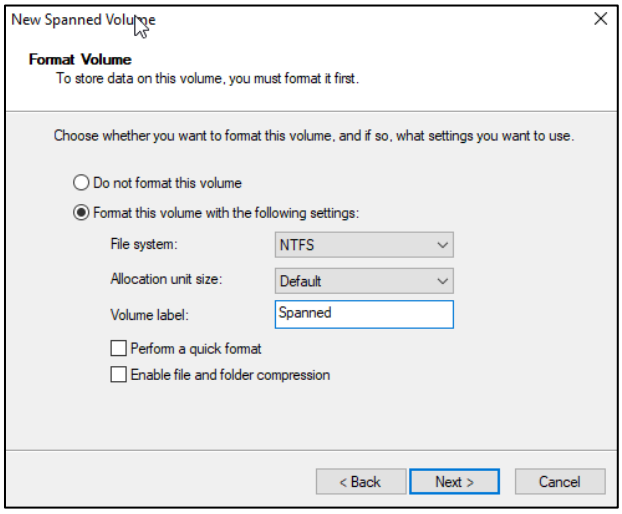
- Now the next screen will be appear again Press ***Next.***



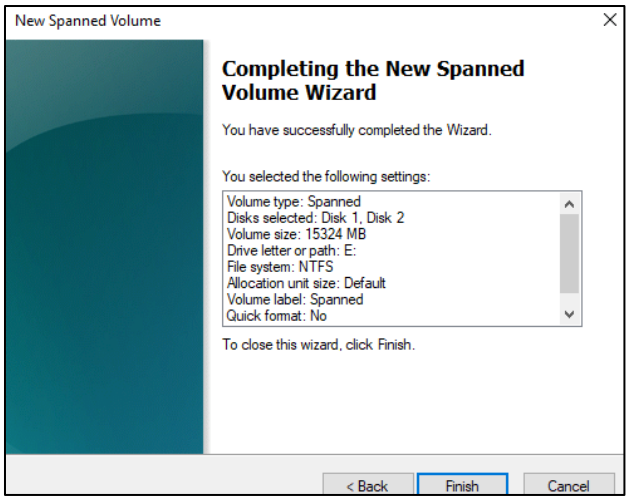
- Next screen will be.



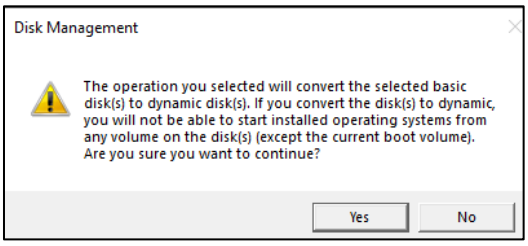
- Change the name of ***Volume label to spanned*** & press ***Next***.



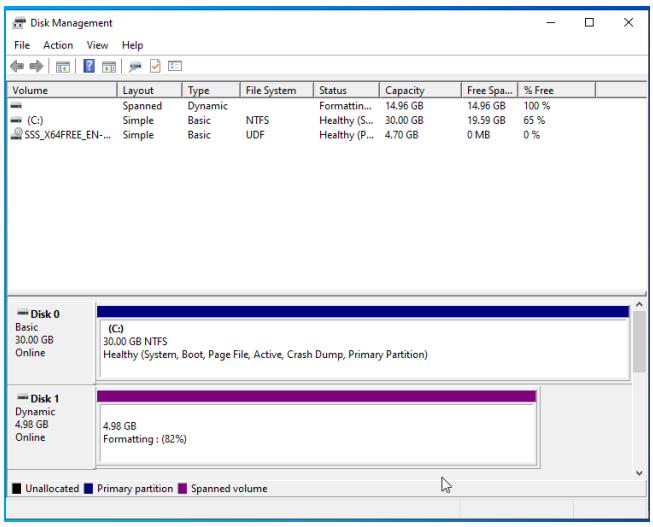
- Press ***Finish*** button show below.



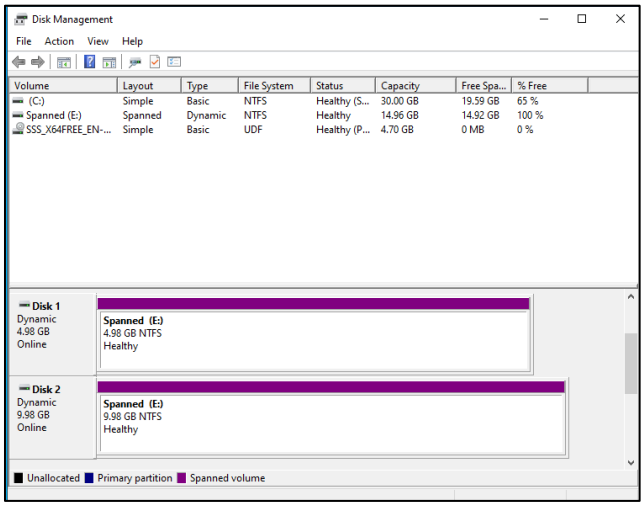
- Press ***Yes*** for this message.



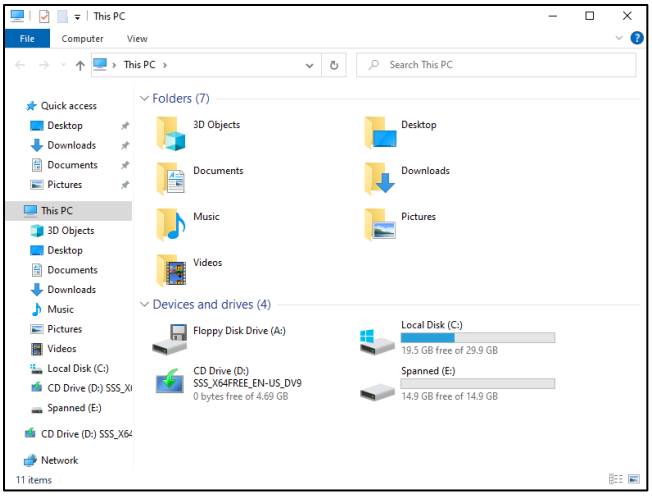
- In disk management see ***color of disk 1 & disk 2 change***. Both disks formatting in process wait for complete the formatting.



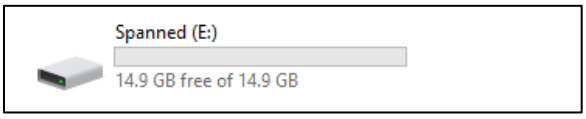
- The screen show the ***spanned method*** successfully completed.



- Then move to ***This PC.***

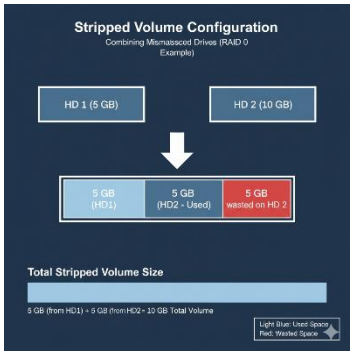


- There is ***spanned (E:) drive of 15GB***. Disk 1 is 5GB & Disk 2 is 10GB so the new ***spanned volume*** merge both 10+5=15GB.



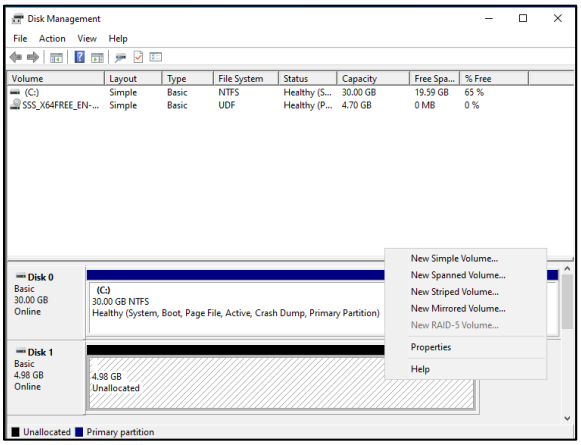
ii) Stripped Volume (Working):

Feature	Details
Configuration Name	Striped Volume (RAID 0)
Primary Goal	Achieve maximum speed by parallel access (high read/write performance).
How It Works	1. Data is broken into tiny pieces (stripes). 2. These pieces are written simultaneously across all disks in the array.
Capacity Requirement	Requires equal space from every disk to function optimally.
Capacity Example	HD 1 (5 GB) & HD 2 (10 GB)
Total Size (Example)	5 GB + 5 GB = 10 GB
Drawback	Zero protection (if either the 5 GB disk or the utilized part of the 10 GB disk fails, the entire 10 GB volume is lost).

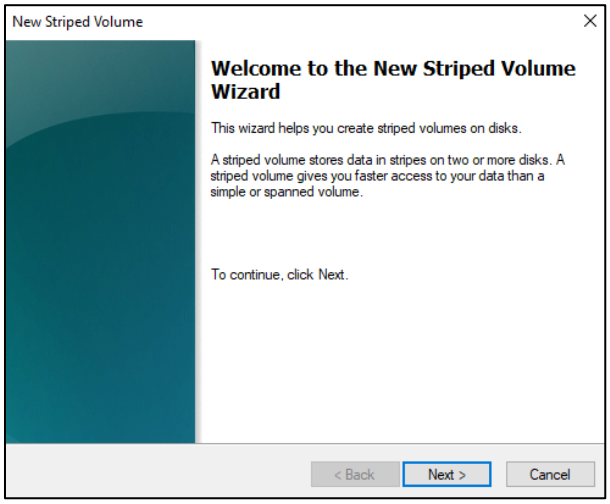


Apply Stripped Volume:

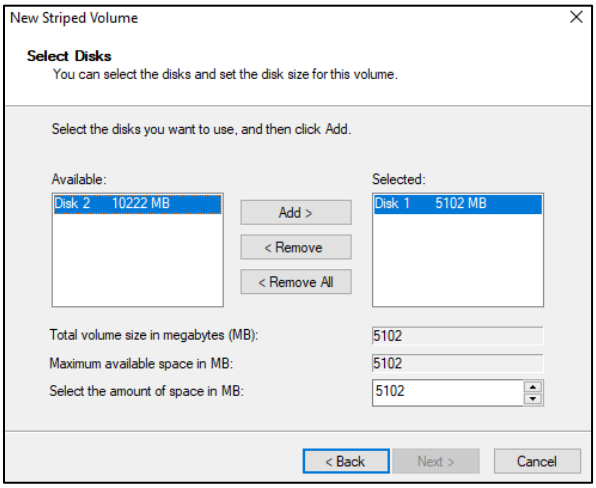
- Again open disk management first ***unallocated the disks*** by deleting volume. ***Right Click on Disk 1***, then different options appears on screen. ***Choose New Stripped Volume.***



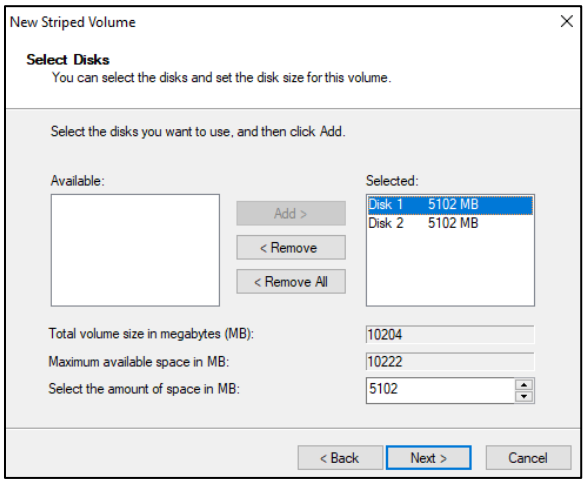
- Then this screen will be appear press ***Next***.



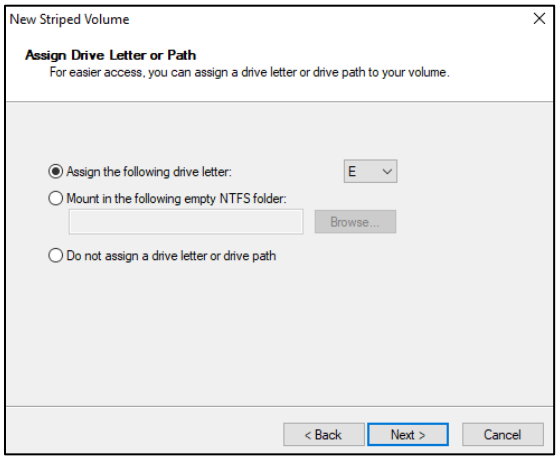
- Then this screen will be appear ***Disk 1 on Right*** & ***Disk 2 on left*** select disk 2 & press ***ADD***.



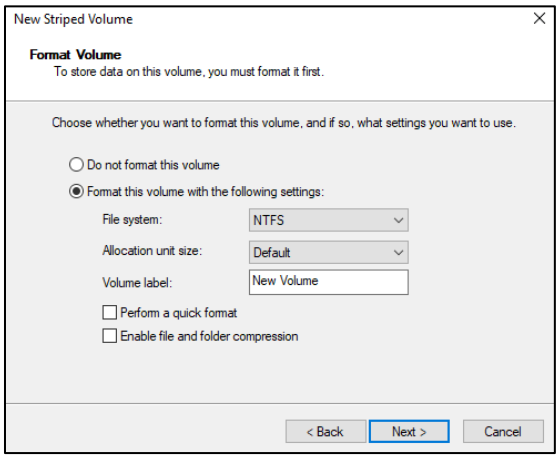
- ***The Disk 2 is now in selected box below Disk 1***. Press **Next**.



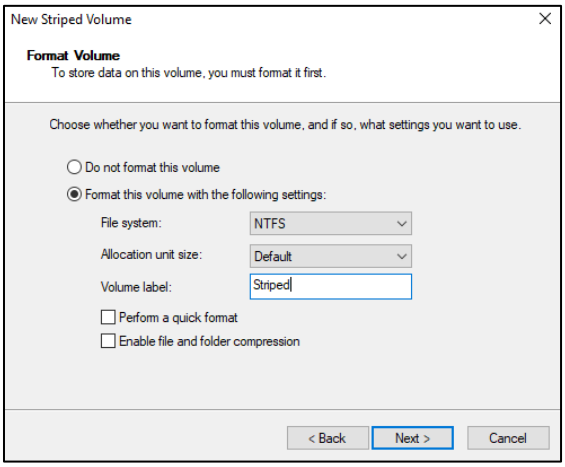
- Then the screen appears Press ***Next.***



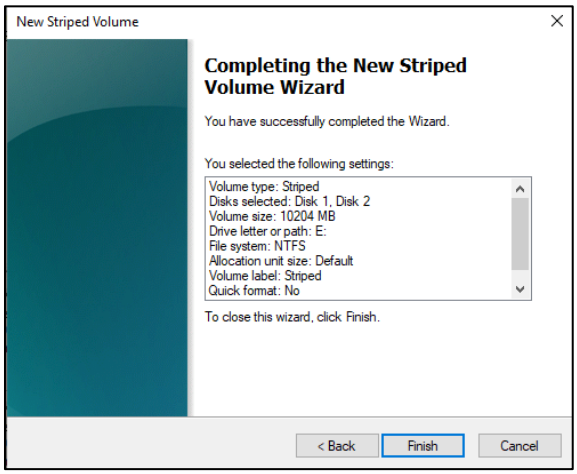
- Next this screen will be appears.



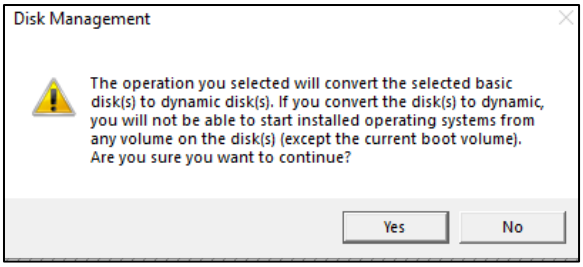
- Change the name of ***Volume label to Striped*** & press ***Next***.



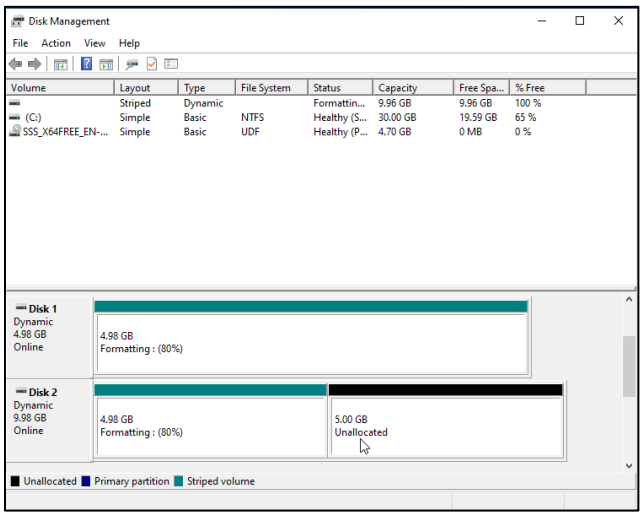
- Press ***Finish*** button show below.



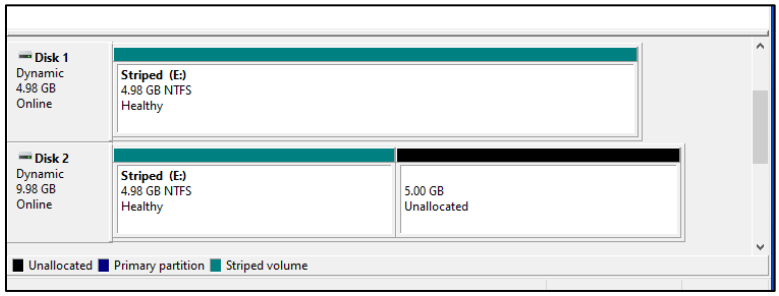
- Press ***Yes*** for this message.



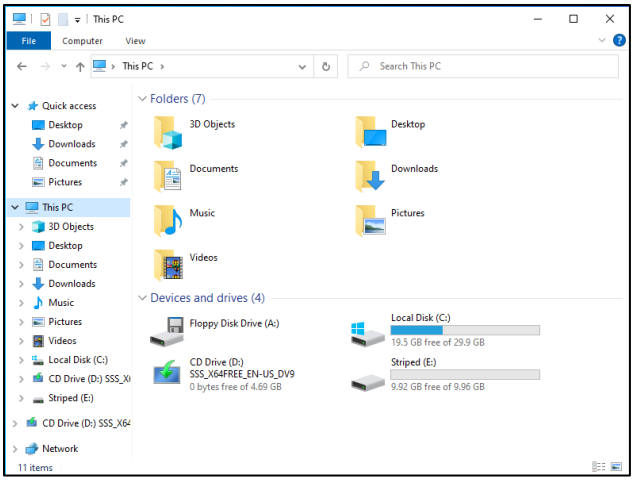
- In disk management see ***color of disk 1 & disk 2 change***. Both disks formatting in process wait for complete the formatting.



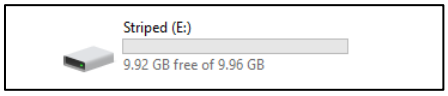
- The screen show the ***Striped method*** successfully completed.



- Now move to ***This PC.***



- There is ***Stripped (E:) drive of 10GB***. Disk 1 is 5GB & Disk 2 is 10GB so the new ***Stripped volume*** merge both 5+5=10GB & 5GB for data protection.



♦ RAID 1:

RAID 1 (Mirrored Volume): The Data Safety Method

Focuses on **data protection** by creating an exact duplicate (mirror) of your data.

- **How it Works:** Data is identically copied to both disks, creating an exact duplicate.
- **Limitation:** Requires at least two disks and provides only the capacity of the smallest disk.
- **Benefit:** Fault tolerance; if one disk fails, the data is preserved on the other disk, and the system can continue operating.
- **Goal:** High data redundancy and availability.

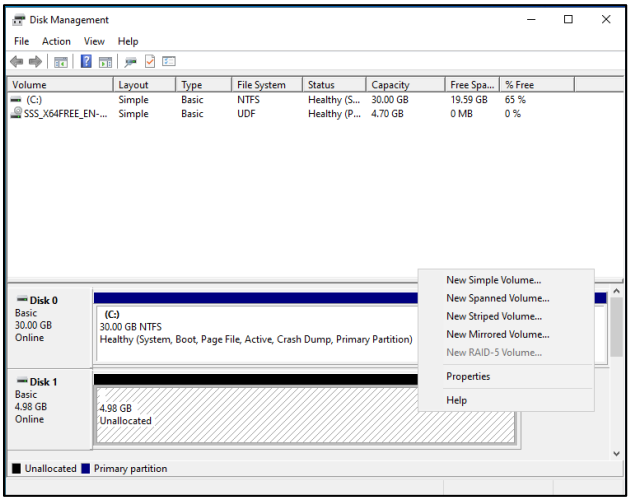
Implementation with Disks not equal (5 GB & 10 GB Example)

When combining a **5GB** & **10GB** disks into a RAID 1 array:

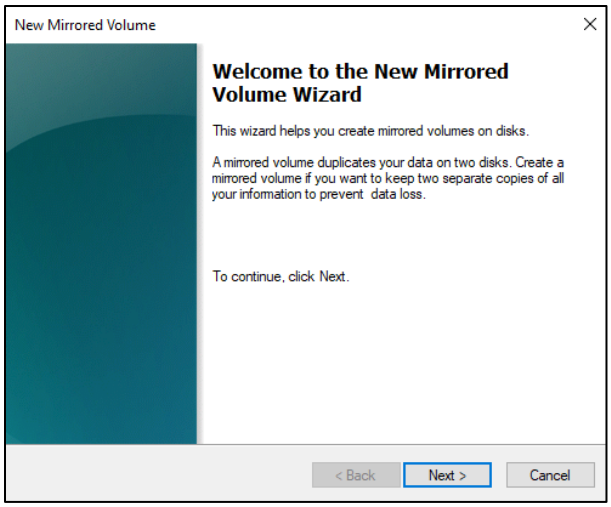
Disk Setup	Capacity Utilization	Resulting Volume
HD 1 (Primary)	**5 GB** (Fully utilized)	**5 GB** (Total Usable Space)

HD 2 (Mirror)	**5 GB** (Utilized to mirror HD 1)	
Unused Space	5 GB(Remaining on HD 2, not used for mirroring)	

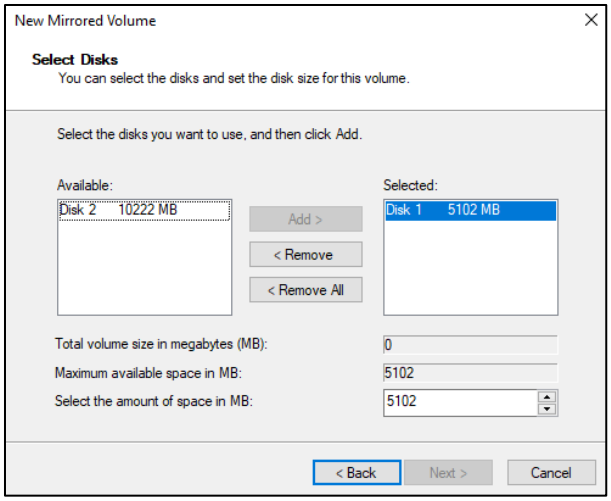
Apply Mirrored Volume: Again open disk management first unallocated the disks by deleting volume. ***Right Click the Disk 1***, then different options will be appear. ***Choose New Mirrored Volume.***



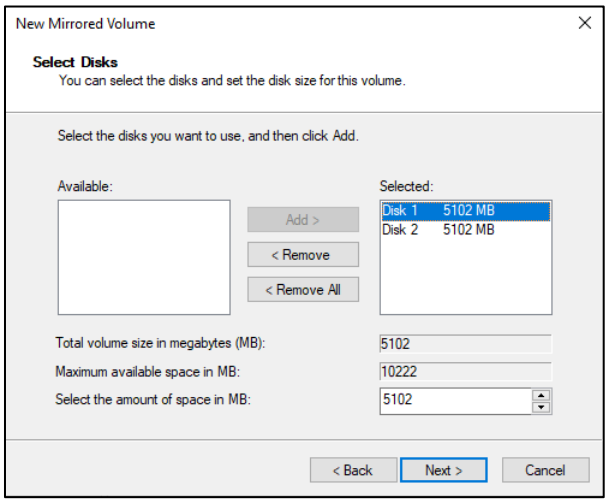
- Then this screen will appear Press ***Next***.



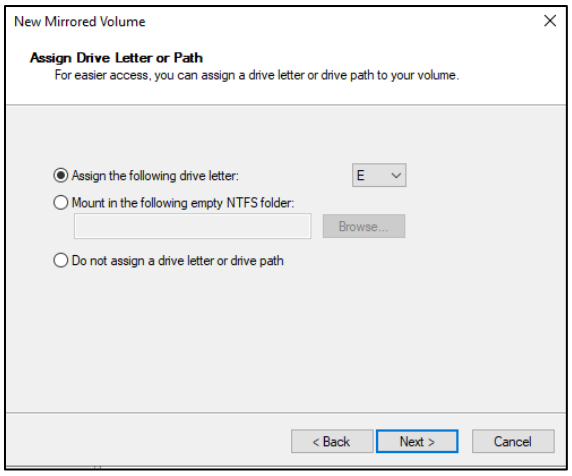
- Then this screen will be appear ***Disk 1 on Right*** & ***Disk 2 on left*** select disk 2 & press ***ADD***.



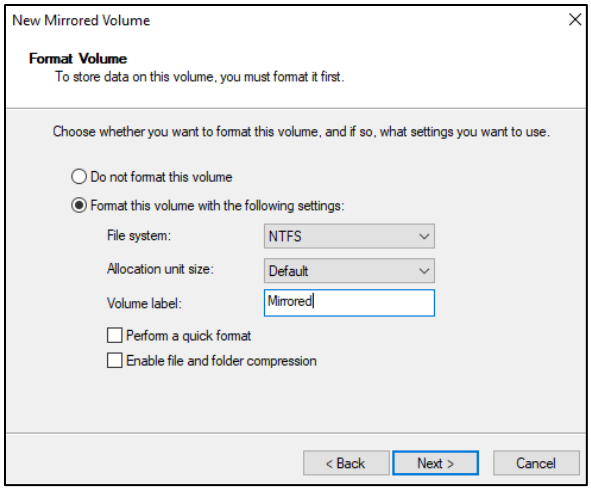
- ***The Disk 2 is now in selected box below Disk 1***. Press **Next**.



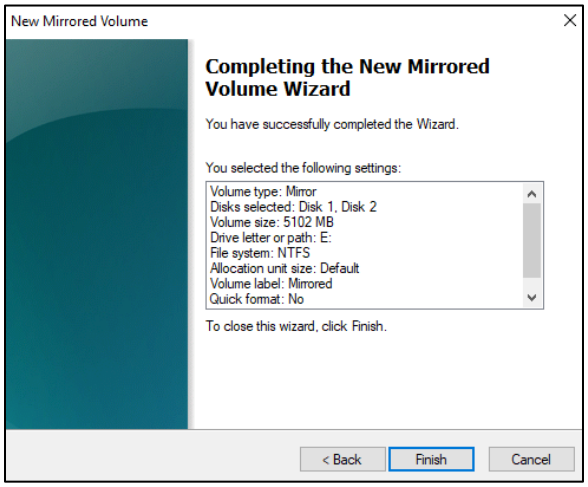
- Then the screen appears Press ***Next.***



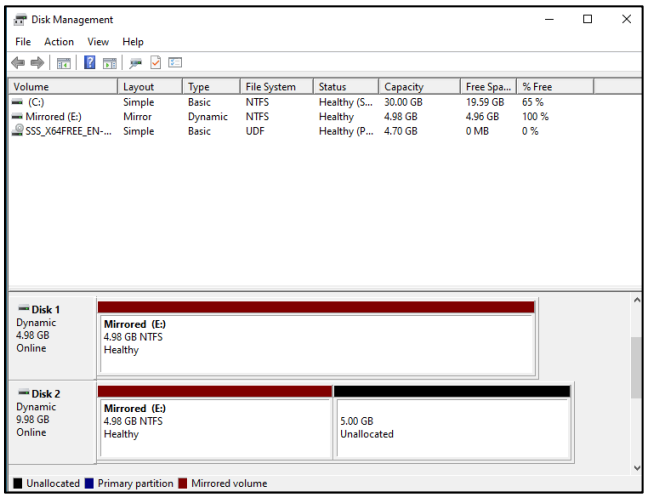
- Change the name of ***Volume label to Mirrored*** & press ***Next***.



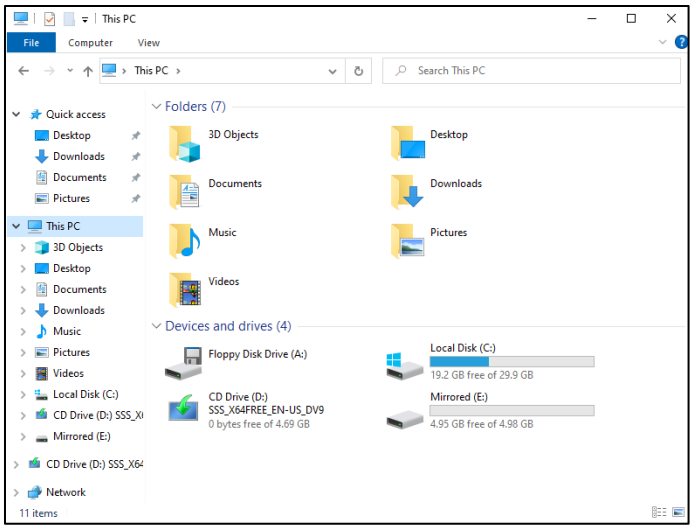
- Press ***Finish*** button show below.



- In disk management ***color of disk 1 & also disk 2 changes***. The Mirrored method completed successfully.



- Then move to ***This PC.*** Now there is ***Mirrored (E:) drive of 5GB***. ***Disk 1 is 5GB*** & ***Disk 2 is 10GB*** due to mirroring it will be 5GB, the new Mirrored volume is mirror/reflection of Disk 1 (5 GB) & ***5GB is for data protection***.



♦ RAID 5: The "Balanced" Method (Working)

Feature	Details
Configuration Name	RAID 5: The "Balanced" Method
Minimum Disks	3 disks
Primary Goals	Balance of speed (striping), capacity, and protection (parity).
How It Works	Data is striped across all disks, and a distributed parity block is used as a recovery key.
Protection	Can withstand the failure of any single disk without data loss.
Usable Capacity	Total capacity equals the size of all disks minus one disk's worth of space (used for distributed parity).
Capacity Formula	(Number of Disks - 1) * (Size of Smallest Disk)

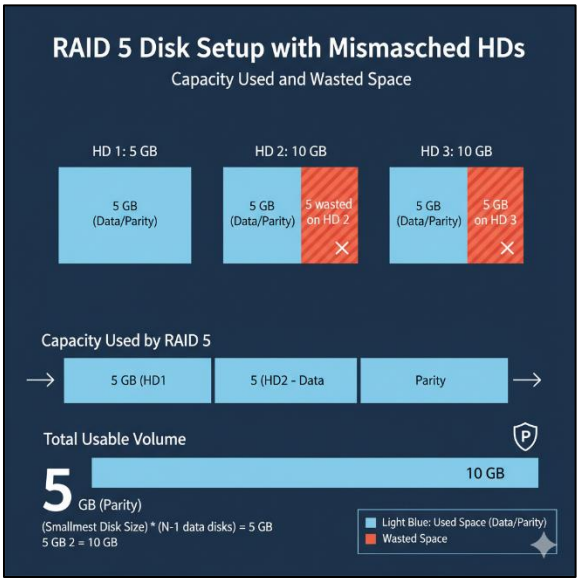
Implementation with Unequal Disks (5 GB, 10 GB, 10 GB Example)

When combining three disks of unequal size in RAID 5, the capacity is governed by the **smallest disk (5 GB)**.

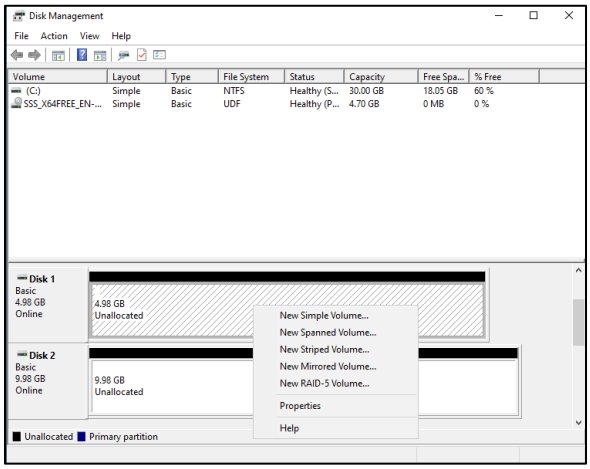
- **Capacity Used:** The array uses **5 GB** from **each** of the three drives.
- **Parity Cost:** One 5 GB block is reserved for parity.
- **Usable Space:** The total usable volume is 10GB (two disks' worth of space).

- **Wasted Space:** The remaining 5GB on the 10GB disks is **unused** by the array, as the array must maintain the smallest disk's size across all members.

Disk Setup	***Capacity Used by RAID 5***	***Total Usable Volume***
HD 1: *5 GB*	**5 GB** (Data/Parity)	10 GB
HD 2: *10 GB*	**5 GB** (Data/Parity)	5 GB wasted on HD 2
HD 3: *10 GB*	**5 GB** (Data/Parity)	5 GB wasted on HD 3

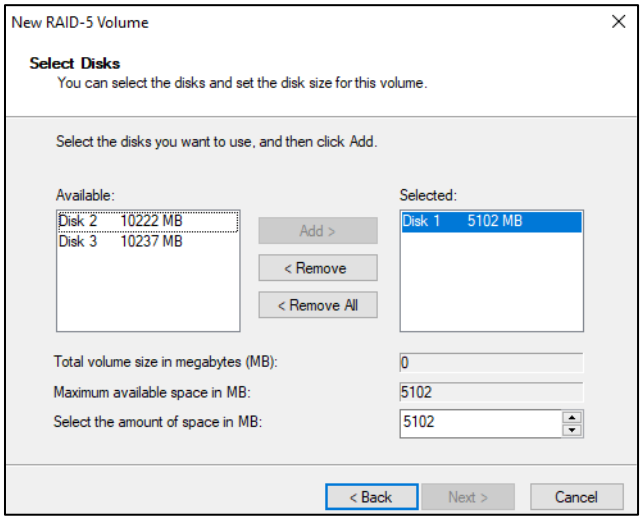


- Again move to disk management ***first unallocated the disk 1*** by deleting volume. ***Right Click the Disk 1***, then different options will be appear. See ***RAID-5 is visible*** because we have three disks. Press ***New RAID-5 Volume.***

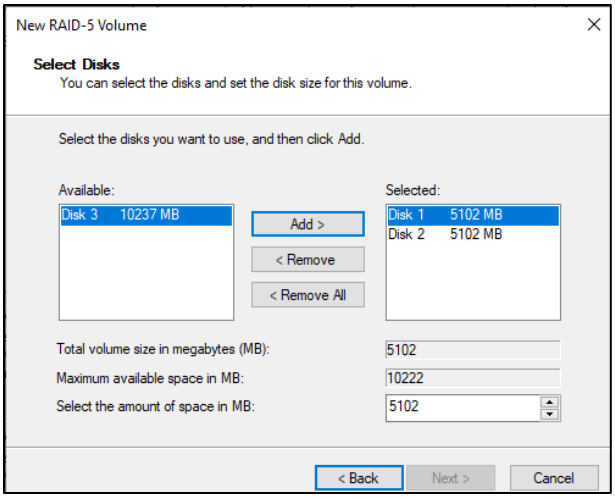


- Then this screen will appear Press ***Next***.

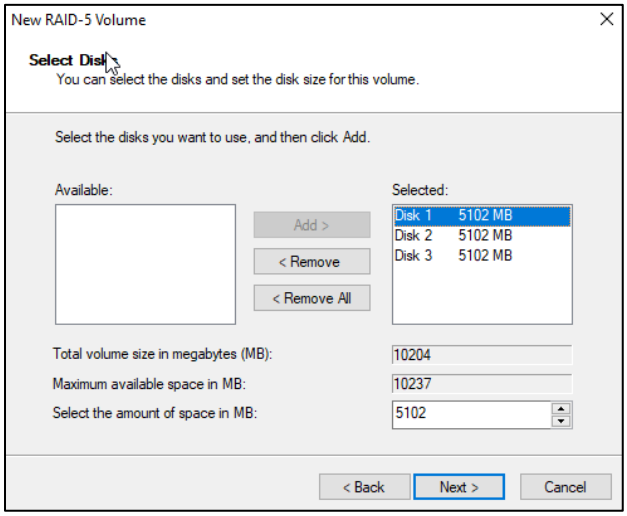
- Then this screen will be appear ***Disk 1 on Right*** & ***Disk 2 & Disk 3 on left***.



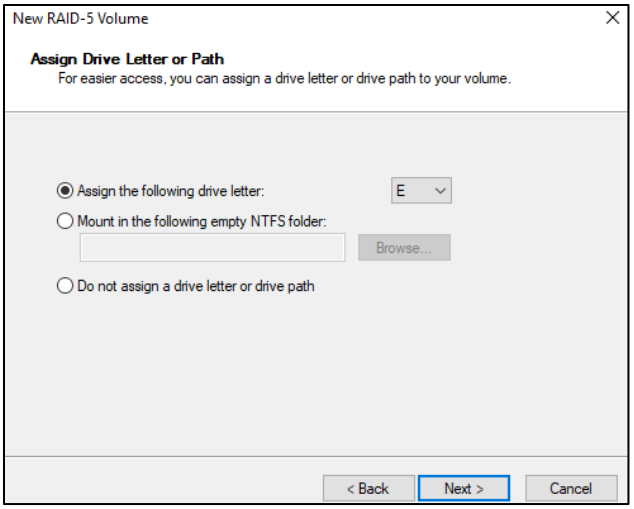
- Now ***select Disk 2 then Press ADD***. ***Disk 2 is now showing in selected box below Disk 1***.



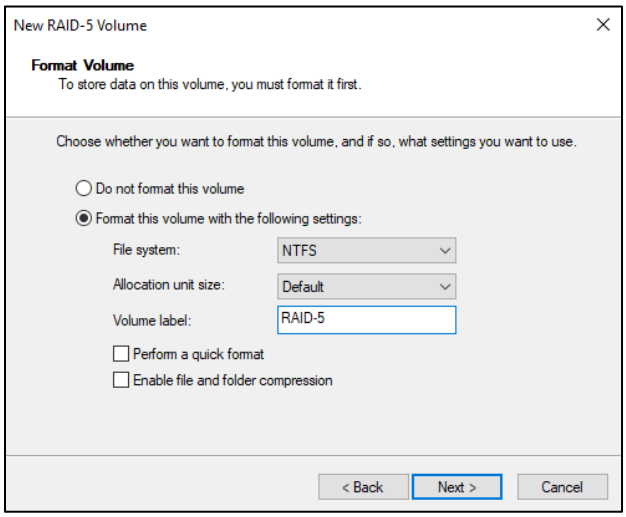
- Now ***select Disk 3 then Press ADD. Disk 3 is now showing in selected box below Disk 2***. Then Press ***Next***.



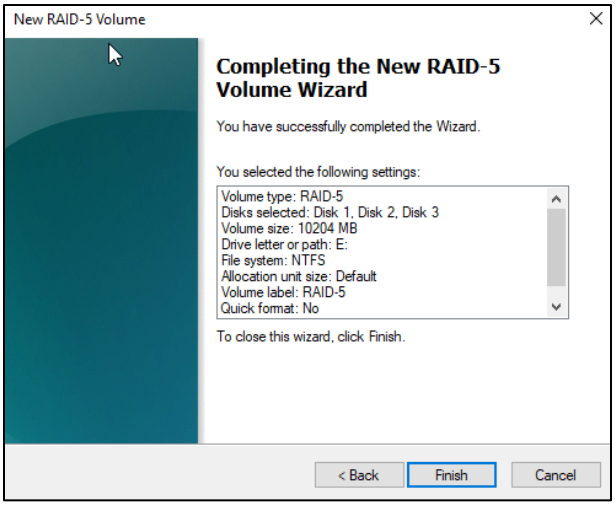
- Then the screen appears Press ***Next.***



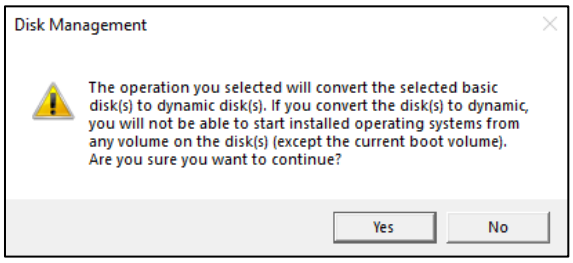
- Change the name of ***Volume label to RAID-5*** & press ***Next***.



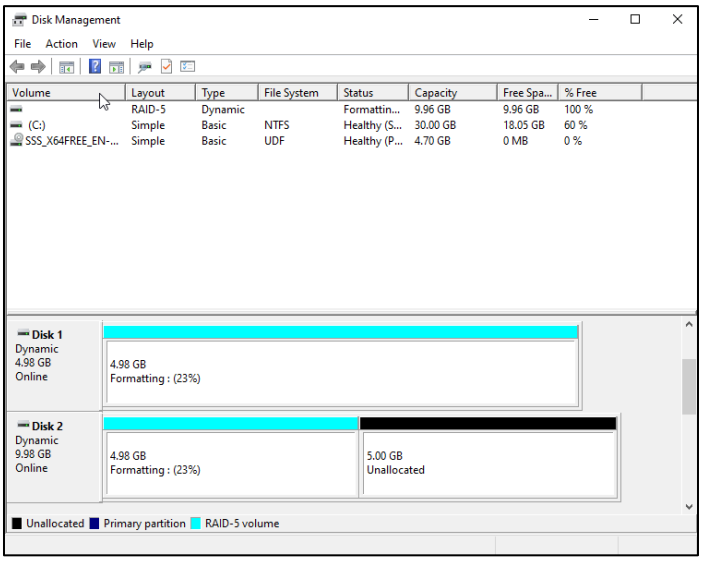
- Press ***Finish*** button show below.



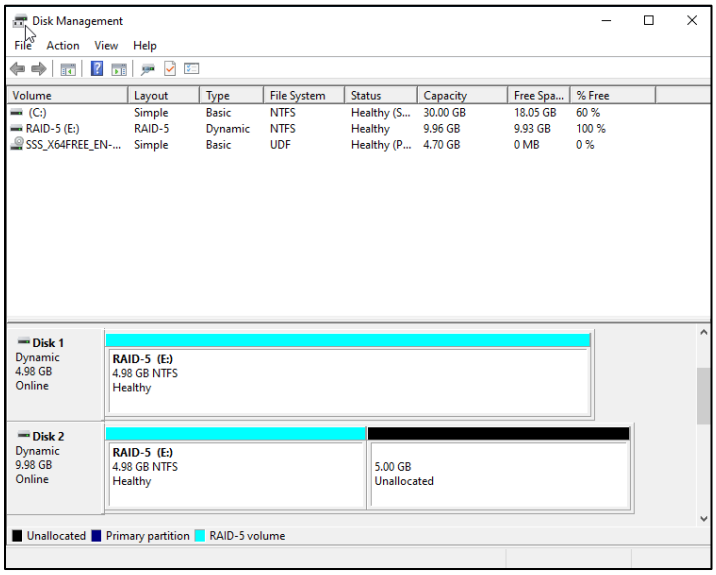
- Press ***Yes*** for this message.



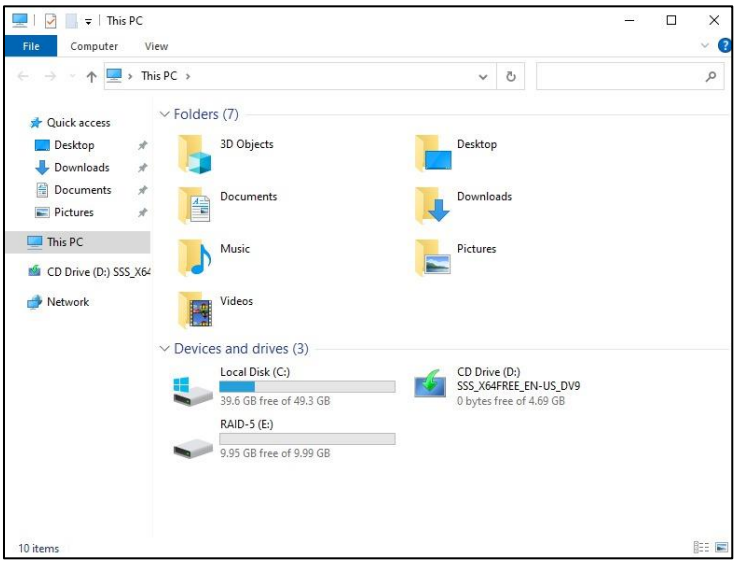
- In disk management ***color of disk 1, disk 2 & also disk 3 changes***. All disks formatting in process wait for complete the formatting.



- The RAID-5 method completed successfully.



- Now ***RAID disk of 10GB***.



☆ THE END! ☆